

Deliverable 3 Frontend and AI Chatbot Test Cases

1. Scope

This document defines the comprehensive manual frontend test suite for Deliverable 3.

In scope:

- all user-facing UI routes in the Next.js frontend
- all frontend flows for UC1 to UC7
- the distinguishable Deliverable 3 feature: PrimeBid AI chatbot and auto-bid assistant
- client-side validation, navigation, usability, responsive behavior, and negative-path behavior

Out of scope:

- backend unit tests and backend implementation changes
- backend load-test implementation details
- database-level validation already covered by backend testing

2. Frontend Routes Under Test

Route	Page	Primary coverage
/	Home	landing page, navigation entry points
/signup	Sign up	UC1 registration
/login	Login	UC1 authentication
/catalogue	Catalogue	UC2 browse and search
/catalogue/[id]	Item detail	UC2, UC3, UC4, AI chatbot
/dashboard/sell	Sell item	UC7 create item and auction
/dashboard/settings	Security settings	password reset
/payment/[id]	Payment page	UC5 payment
/receipt/[id]	Receipt page	UC6 receipt

3. Test Environment

- Frontend running at `http://localhost:3000`
- Gateway running at `http://localhost:8080`
- IAM, catalogue, auction, and payment services reachable through the gateway
- Docker Compose or local service startup already completed
- `GEMINI_API_KEY` configured if live AI responses are being tested

Recommended test accounts:

- `seller_a3`
- `buyer_a3`
- `buyer_competitor_a3`

Recommended reusable test data:

- one active auction item
- one expired item with no bids
- one sold item with the logged-in user as winner
- one sold item with the logged-in user as non-winner

3.1 Test Data Preparation

Before executing this suite, prepare the necessary system state using the Deliverable 2 API walkthroughs:

- [INDIVIDUAL_USE_CASE_TESTING.md](#)
- [INDIVIDUAL_USE_CASE_TESTING_MAC.md](#)

Preparation expectations:

- create users for seller, buyer, and competing buyer flows
- create at least one active auction item
- create one sold item where the logged-in buyer is the winner
- create one sold item where the logged-in buyer is not the winner
- create one ended auction with no bids

3.2 Viewports Used During Manual Testing

Responsive checks in this suite use:

- mobile viewport: `390 x 844`

- desktop viewport: 1440 x 900

3.3 What This Test Suite Means for Frontend Testing

- verify that pages render correctly
- verify that navigation and user flows work
- verify that client-side validation blocks bad input
- verify that backend failures surface safely in the interface
- verify that the AI assistant behaves correctly from the user's point of view

For this deliverable, frontend verification is documented as a manual functional test suite:

- the tester executes the steps directly in the UI
- the tester confirms the expected result on-screen
- the suite records both happy paths and edge cases

For the AI chatbot, exact wording is not the right assertion target. AI validation should be based on:

- whether the assistant returns a usable answer
- whether it recommends a valid action when appropriate
- whether it respects budget and auction state constraints
- whether the UI renders success, error, and fallback states safely

AI-specific note:

- AI-04 through AI-14 assume a valid GEMINI_API_KEY and a working gateway AI integration path
- without a live key, only shell behavior such as assistant visibility, layout, and disabled-state handling can be verified

3.4 Edge-Case Coverage Included in This Suite

This suite intentionally includes edge cases, not only happy paths. The covered edge-case categories include:

- duplicate signup and invalid login
- missing required auth, seller, bidding, and payment inputs
- unauthorized access to seller-only or winner-only pages
- invalid bid amounts and ended-auction bidding attempts
- invalid payment expiry dates and payment access by non-winners
- seller misuse of the AI auto-bid flow
- low-budget, duplicate-session, ended-auction, and degraded-response AI scenarios

4. Traceability Matrix

Use case / feature	Frontend pages	Test IDs
UC1 Registration and Login	/signup , /login , /dashboard/settings , navbar auth controls	NAV-01 to NAV-04 , AUTH-01 to AUTH-10 , SET-01 to SET-04
UC2 Browse, Search, View, Select Item	/ , /catalogue , /catalogue/[id]	HOME-01 to HOME-03 , CAT-01 to CAT-08
UC3 Place Bid and View Bid History	/catalogue/[id]	BID-01 to BID-07
UC4 Auction Result	/catalogue/[id]	AUC-01 to AUC-05
UC5 Payment	/payment/[id]	PAY-01 to PAY-09
UC6 Receipt	/receipt/[id]	REC-01 to REC-05
UC7 Create Item and Auction	/dashboard/sell , /catalogue , /catalogue/[id]	SELL-01 to SELL-08
A3 Distinguishable Feature: PrimeBid AI	/catalogue/[id]	AI-01 to AI-14

5. Shared Navigation and Layout Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
NAV-01	High	Verify landing page entry points	User is logged out	Open /	Home page loads without console-breaking UI errors; primary CTA goes to catalogue; secondary CTA goes to signup
NAV-02	High	Verify auth navbar while logged out	User is logged out	Open /catalogue on desktop and mobile widths	Navbar shows Catalogue , Sign in , and Create account ; mobile menu opens and closes correctly
NAV-03	High	Verify auth navbar while logged in	User is logged in	Open /catalogue	Navbar shows Sell Item , Settings , user identity chip, and sign-out control

ID	Priority	Objective	Preconditions	Steps	Expected result
N A V - 0 4	Medium	Verify auth pages hide the shared navbar	User is logged out	Open <code>/login</code> and <code>/signup</code>	Global navbar is hidden on auth pages; only the auth page menu container is visible

6. Home Page Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
HOM E- 01	High	Verify home page content renders	None	Open <code>/</code>	Hero title, CTA buttons, and feature cards render correctly
HOM E- 02	High	Verify route to catalogue from home	None	Click <code>Enter Catalogue</code>	User is routed to <code>/catalogue</code>
HOM E- 03	High	Verify route to signup from home	None	Click <code>Create account</code>	User is routed to <code>/signup</code>

7. UC1 Authentication Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
A UT H- 0 1	High	Register a new user successfully	Unique username available	Open <code>/signup</code> , complete all fields, submit	User is redirected to <code>/login?registered=true</code> ; success banner appears
A UT H- 0 2	High	Validate required signup fields	None	Submit signup form with one or more blank fields	Browser blocks submission or UI prevents submit until required fields are provided
A UT H- 0 3	High	Validate minimum signup password length	None	Enter password shorter than 6 characters and submit	Submission is blocked or validation message appears

ID	Priority	Objective	Preconditions	Steps	Expected result
A UT H- 04	High	Handle duplicate username on signup	Existing username already present	Submit signup form with an existing username	Error banner appears and the page stays usable
A UT H- 05	High	Login successfully with valid credentials	Registered user exists	Open <code>/login</code> , enter valid credentials, submit	User is routed to <code>/catalogue</code> ; authenticated navbar state is shown
A UT H- 06	High	Reject invalid login credentials	Registered user exists	Attempt login with wrong password	Error banner appears; user remains on login page
A UT H- 07	High	Prevent login submit with empty fields	None	Leave username or password blank and attempt submit	Submit button remains disabled or browser blocks submission
A UT H- 08	Medium	Preserve post-signup feedback	A user has just registered	Land on <code>/login?registered=true</code>	Success state is visible and instructs the user to sign in
A UT H- 09	High	Verify logout behavior	Logged-in user session exists	Click sign-out from navbar or mobile menu	Session clears and navbar returns to logged-out state
A UT H- 10	High	Prevent unauthorized privileged actions	User is logged out	Attempt seller-only or payment-only action from UI	Action does not complete; UI shows error or redirects to sign in

8. Settings and Password Reset Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
S E T- 01	Medium	Load security settings page	Logged-in user	Open <code>/dashboard/settings</code>	Security settings page loads; reset password form is visible

ID	Priority	Objective	Preconditions	Steps	Expected result
S E T - 0 2	Medium	Reset password successfully	Logged-in user knows current password	Enter valid current password and valid new password, then submit	Success banner appears; password fields clear
S E T - 0 3	Medium	Reject incorrect current password	Logged-in user	Enter wrong current password and submit	Error banner appears; page stays usable
S E T - 0 4	Medium	Enforce required password-reset inputs	Logged-in user	Leave either field blank and attempt submit	Submit is blocked until both fields are provided

9. UC7 Seller Item Creation Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
S E L L- 0 1	High	Open seller listing page	Logged-in seller	Open <code>/dashboard/sell</code>	Seller form renders with asset details and logistics sections
S E L L- 0 2	High	Create an item successfully	Logged-in seller	Fill all required fields with valid values and submit	User is routed to <code>/catalogue/[itemId]</code> ; item page shows the new listing
S E L L- 0 3	High	Validate required seller form fields	Logged-in seller	Leave a required field blank and attempt submit	Submission is blocked or a visible error is shown
S E L L- 0 4	Medium	Validate numeric constraints on seller form	Logged-in seller	Enter invalid values such as negative price or invalid duration	UI prevents bad input or backend error is surfaced in-page without breaking layout

ID	Priority	Objective	Preconditions	Steps	Expected result
S EL L- 0 5	Medium	Verify create-item CTA status feedback	Logged-in seller	Submit valid form	Button switches to loading state and prevents duplicate submits
S EL L- 0 6	Medium	Verify created item appears in catalogue	Newly created item exists	Return to <i>/catalogue</i>	New item is visible in the list with correct title and status
S EL L- 0 7	Medium	Verify seller item detail information	Newly created item exists	Open the item detail page	Description, condition, keywords, shipping, and seller metadata display correctly
S EL L- 0 8	High	Reject unauthenticated item creation	Logged-out user	Attempt to submit seller form	Item is not created; UI surfaces authorization failure gracefully

10. UC2 Catalogue and Item Detail Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
C AT - 0 1	High	Load catalogue page successfully	None	Open <i>/catalogue</i>	Catalogue title and search UI appear; item cards render if data exists
C AT - 0 2	High	Search catalogue by keyword	At least one matching item exists	Enter search term in the search box	Matching items are shown and irrelevant items are filtered out
C AT - 0 3	Medium	Empty search state handling	Search term has no matches	Search for a non-existent keyword	Empty state message appears without breaking page layout
C AT - 0 4	High	Open item detail page	At least one item exists	Click a catalogue card	User is routed to <i>/catalogue/[id]</i>

ID	Priority	Objective	Preconditions	Steps	Expected result
CAT-05	High	Verify active-auction detail view	Active item exists	Open active item detail page	Current price, timer, seller data, and bid history region display correctly
CAT-06	High	Verify unauthenticated bid prompt	Logged-out user on active item	Open item detail page	Bid area shows sign-in prompt instead of allowing a completed bid
CAT-07	Medium	Verify bid-history rendering	Item has bids	Open item detail page	Bid history shows bids ordered with highest highlighted
CAT-08	Medium	Verify countdown state transition	Active auction close to end	Watch item detail through auction expiry	UI transitions from active to ending/finalizing to ended state without crashing

11. UC3 Bidding Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
BID-01	High	Place a valid bid successfully	Logged-in buyer, active auction, item selected or selectable	Enter a bid above current price and submit	Success message appears; current price and bid history update
BID-02	High	Reject bid below required increment	Logged-in buyer, active auction	Enter a bid below the displayed minimum and submit	Bid is rejected and a clear error message is shown
BID-03	High	Reject empty bid submission	Logged-in buyer, active auction	Leave bid field blank and attempt submit	Submit remains disabled or request is not sent

ID	Priority	Objective	Preconditions	Steps	Expected result
BID-04	Medium	Verify loading state during bidding	Logged-in buyer, active auction	Submit a valid bid	Button enters loading state and duplicate submission is prevented
BID-05	Medium	Verify bid history updates after own bid	Logged-in buyer	Place a bid successfully	New bid appears in bid history and highest state updates
BID-06	Medium	Verify another user can outbid	Two buyers available on same active auction	Buyer A bids, then Buyer B bids higher	Highest bid and leader information update correctly on reload/poll
BID-07	High	Prevent bid attempt after auction ended	Logged-in buyer, ended auction	Attempt to bid after result is final	Bid form is no longer active; user sees ended/result state instead

12. UC4 Auction Result Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
AUC-01	High	Winner sees sold result state	Logged-in winning buyer, sold item	Open ended item detail page	Winner banner appears and payment CTA is visible
AUC-02	High	Non-winner sees sold result state	Logged-in non-winning buyer, sold item	Open ended item detail page	Winner name and final price show; payment CTA is not offered
AUC-03	Medium	Seller sees ended result state	Logged-in seller of the item	Open ended item detail page	Auction is shown as concluded with winner/final result information

ID	Priority	Objective	Preconditions	Steps	Expected result
AUC-04	Medium	Logged-out user sees limited ended state	Logged-out user, ended item	Open ended item detail page	Sign-in prompt is shown for private result details
AUC-05	High	Unsold auction is handled correctly	Ended item with no bids	Open ended item detail page	UI shows Auction Ended or No bids style message instead of payment path

13. UC5 Payment Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
PAY-01	High	Load payment page for winner	Logged-in winner with payable item	Open /payment/[id]	Secure checkout page loads with order summary and shipping address
PAY-02	Medium	Toggle expedited shipping	Logged-in winner on payment page	Toggle expedited shipping on and off	Total updates correctly and toggle state is reflected visually
PAY-03	High	Submit payment with valid details	Logged-in winner on payment page	Enter valid card name, card number, expiry, and CVC, then submit	User is redirected to receipt page
PAY-04	High	Reject past expiry date	Logged-in winner on payment page	Enter an expiry date in the past and submit	Client-side error message appears and payment is blocked
PAY-05	High	Reject invalid expiry month format	Logged-in winner on payment page	Enter month 00 or 13	Validation prevents payment submission

ID	Priority	Objective	Preconditions	Steps	Expected result
PAY-06	High	Enforce required payment fields	Logged-in winner on payment page	Leave one or more payment fields blank and submit	Submission is blocked or page shows validation errors
PAY-07	High	Prevent unauthorized payment access	Logged-in non-winner or logged-out user	Open <code>/payment/[id]</code> for another user's item	Payment cannot be completed; UI surfaces access failure safely
PAY-08	Medium	Verify payment button loading state	Logged-in winner on payment page	Submit valid payment	Button shows processing state and blocks duplicate submits
PAY-09	High	Verify payment page survives backend errors gracefully	Simulate failed payment or unavailable downstream service	Submit payment	Error banner appears and the page remains usable for correction/retry

14. UC6 Receipt Tests

ID	Priority	Objective	Preconditions	Steps	Expected result
REC-01	High	Load receipt page after successful payment	Payment just completed	Follow redirect to <code>/receipt/[paymentId]</code>	<code>Payment Confirmed</code> page appears
REC-02	Medium	Verify receipt financial summary	Successful payment exists	Inspect receipt page	Winning bid, shipping, and total paid values are displayed
REC-03	Medium	Verify masked card and shipping message	Successful payment exists	Inspect receipt metadata card	Cardholder name, masked last four digits, and shipping message are shown

ID	Priority	Objective	Preconditions	Steps	Expected result
R E C - 0 4	High	Verify return path to catalogue	Successful receipt exists	Click <code>Return to Catalogue</code>	User is routed back to <code>/catalogue</code>
R E C - 0 5	Medium	Verify invalid receipt access handling	Unknown or unauthorized receipt id	Open <code>/receipt/[badgeId]</code>	Friendly error state is shown without app crash

15. A3 PrimeBid AI Chatbot and Auto-Bid Tests

Important note:

- expected AI wording should be evaluated semantically, not by exact sentence match
- the frontend should be judged on response rendering, action affordances, validation, and failure handling

ID	Priority	Objective	Preconditions	Steps	Expected result
A I - 0 1	High	AI assistant is hidden for logged-out users	Logged-out user on item detail page	Open an active item detail page	PrimeBid AI panel is not shown
A I - 0 2	High	Open AI assistant as authenticated buyer	Logged-in buyer on active item	Click <code>PrimeBid AI</code>	Assistant expands with welcome message, budget field, prompts, and input box
A I - 0 3	Medium	Close AI assistant cleanly	Logged-in buyer on active item	Open assistant, then click <code>Close AI Assistant</code>	Panel collapses without affecting the rest of the page
A I - 0 4	High	Ask for auction analysis	Logged-in buyer on active item; live Gemini key configured	Open assistant, send a message asking for analysis	AI response renders in chat and stays within the UI bubble layout

ID	Priority	Objective	Preconditions	Steps	Expected result
A I - 0 5	High	Ask for strategy recommendation with budget	Logged-in buyer on active item; live Gemini key configured	Enter a budget, ask AI how to bid	AI returns a recommendation and may include suggested strategy/max bid
A I - 0 6	High	Start auto-bid suggestion flow	Logged-in buyer on active item with budget entered; live Gemini key configured	Ask AI to start auto-bidding	UI renders an <i>Activate ...</i> action button when AI returns <i>START_AUTOBID</i>
A I - 0 7	High	Activate AI auto-bidding	Logged-in buyer on active item	Click the AI activation button	Auto-bid session status bar appears with strategy, bid count, and stop control
A I - 0 8	Medium	Poll live auto-bid status	Active auto-bid session exists	Leave assistant open during auction	Status updates without page crash; action log state continues to refresh
A I - 0 9	High	Stop AI auto-bidding	Active auto-bid session exists	Click <i>Stop</i> in the assistant	Session stops; assistant shows confirmation message
A I - 1 0	High	Reject seller activating AI on own item	Logged-in seller on own item detail page	Attempt to activate auto-bidder via AI assistant	UI shows readable error returned from backend; no active session bar remains
A I - 1 1	High	Reject auto-bid when budget is below current price	Logged-in buyer on active item	Attempt AI activation with a too-low budget	UI shows an error and does not create a session
A I - 1 2	High	Reject duplicate active auto-bid sessions	Logged-in buyer with one active auto-bid session	Try to activate a second session for the same item	UI shows duplicate-session error rather than creating another active session
A I - 1 3	High	Disable AI chat after auction ends	Logged-in buyer on ended item	Open assistant after end state	Input is disabled or placeholder indicates auction has ended

ID	Priority	Objective	Preconditions	Steps	Expected result
AI-14	High	AI error handling / fallback rendering	Force AI request failure or fallback path; live Gemini path configured	Send an AI message during degraded conditions	Assistant shows a safe error/fallback response in chat and the page remains usable

16. Mobile and Responsive Checks

These checks should be run at minimum on the mobile viewport 390 x 844 and the desktop viewport 1440 x 900 .

ID	Priority	Objective	Preconditions	Steps	Expected result
RESP-01	High	Auth menu remains readable on mobile	Logged-out user	Open <code>/login</code> and <code>/signup</code> on mobile width	Auth menu container remains visible, readable, and non-overlapping
RESP-02	High	Mobile navbar drawer works	Logged-in and logged-out sessions	Open mobile menu on catalogue	Drawer opens/closes and routes remain tappable
RESP-03	High	Signup form stacks correctly on mobile	Logged-out user	Open <code>/signup</code> on mobile width	Name fields stack cleanly and no content clips off-screen
RESP-04	Medium	Sell form remains usable on mobile	Logged-in seller	Open <code>/dashboard/sell</code> on mobile width	All required inputs remain visible and actionable without layout breakage
RESP-05	High	Payment form remains usable on mobile	Logged-in winner	Open payment page on mobile width	Form fields, toggle, totals, and CTA remain readable and tappable

17. Exit Criteria

The frontend and AI chatbot testing pass is considered complete when:

- all high-priority tests in UC1 to UC7 pass
- all high-priority AI tests AI-01 to AI-14 pass
- no blocking navigation, authentication, bidding, payment, or receipt regressions remain
- mobile checks complete for auth, catalogue, sell, and payment pages

- known issues are documented with reproducible steps

18. Execution Summary

This document serves as the Deliverable 3 frontend and AI testing record.

Summary of execution:

- all primary frontend flows for UC1 to UC7 were exercised
- the PrimeBid AI assistant UI, action flow, and edge-case handling were verified
- validation cases for authentication, bidding, payment, and receipt behavior were checked
- responsive checks were completed on both mobile and desktop viewport sizes